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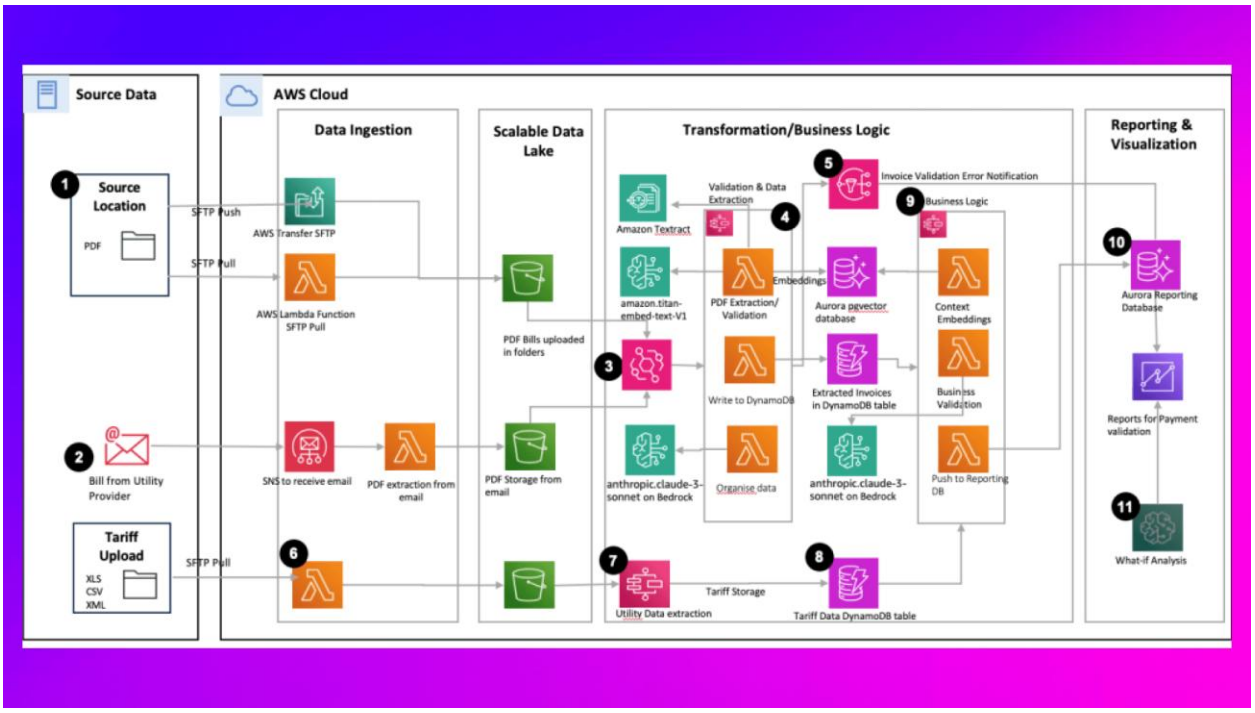
Amazon Bedrock & SageMaker

- ## Amazon Bedrock



Amazon Bedrock is a serverless, fully managed platform that makes generative AI easier to use. Without requiring significant machine learning knowledge, it enables developers to swiftly access robust foundation models from leading AI firms like Anthropic, Stability AI, and Amazon itself. Bedrock provides a quick and simple approach to include these features, whether you're developing applications for text production, image creation, or other AI-driven tasks.

Infrastructure management is not a concern when using Bedrock. Because of its serverless architecture, which automatically scales to meet your needs, you can concentrate on developing cutting-edge apps. The platform's low-code interface makes it usable by companies of all kinds, from startups to large corporations, with the goal of swiftly and effectively implementing AI.

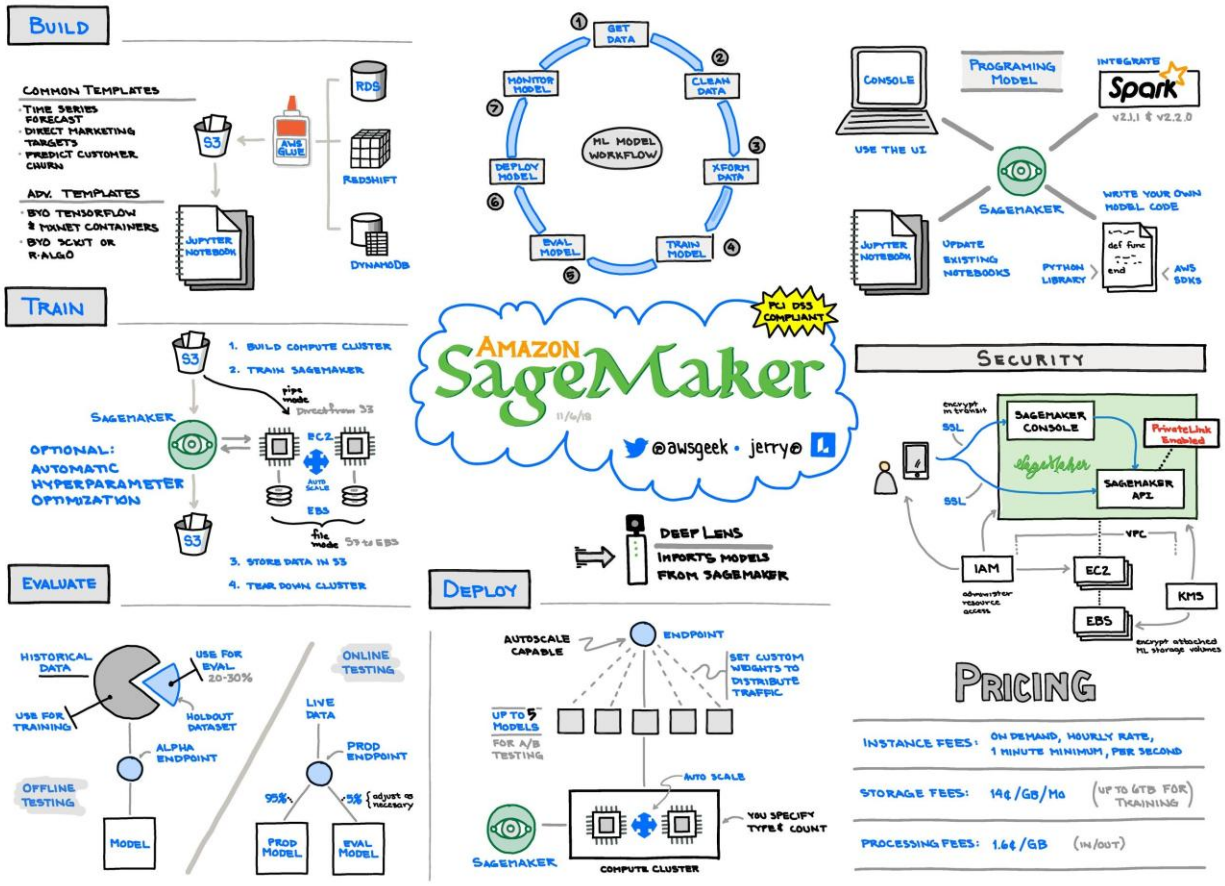
Key Features

1. Pre-trained Foundation Models
2. Serverless Architecture
3. Flexible API Integration
4. Security and Compliance

Core Components

1. Foundation Models
2. Serverless Compute
3. API Layer

- ## Amazon SageMaker



Everything you need to create, train, and implement machine learning models is available on the all-inclusive Amazon SageMaker platform. For individuals who wish to customise and oversee every facet of the machine learning process, SageMaker is a potent alternative to Bedrock, which concentrates on pre-trained models. SageMaker offers the versatility to satisfy a wide range of use cases, whether you're using pre-trained models or training your own.

From data preparation to model training and deployment, SageMaker gives you complete control over the machine learning lifecycle. It offers tools and services that facilitate large-scale, production-ready AI applications and has a tight integration with the AWS environment.

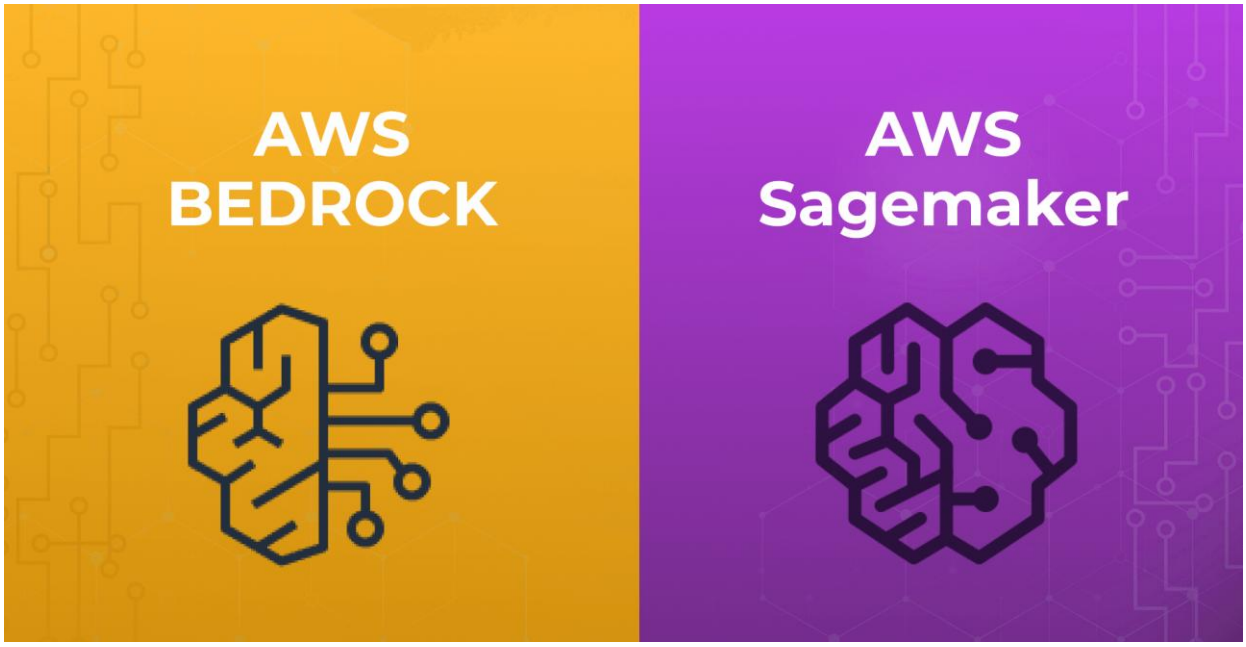
Key Features and Capabilities

- **End-to-End ML Lifecycle**
- **Model Hosting**
- **Comprehensive Toolkit**
- **AWS Integration**
- **Security and Compliance**

Core Components

- **SageMaker Studio**
- **SageMaker Notebooks**
- **Training and Tuning Services**
- **Model Deployment Options**

Comparison



Feature Category	Amazon Bedrock	Amazon SageMaker
Model Access	Pre-trained Foundation Models	Custom Training & Marketplace Models
Deployment Options	Serverless (API-driven)	Instance-based & Serverless
Development Experience	API-first, Low-code	Full ML Development Lifecycle (Code-first, UI, and Notebooks)
Cost Model	Pay-per-use (based on usage)	Instance + Usage Based (for training, inference)
Integration Capabilities	Integrates with AWS services	Integrates with AWS + External Tools (e.g., TensorFlow, PyTorch)
Startup Time	Immediate (no setup needed)	Minutes (Instance setup is required)
Auto-scaling	Built-in, automatically adjusts	Configurable (user-defined settings)
Max Concurrent Requests	Specific limit based on the model	Instance dependent (varies based on instance type)
Regional Availability	Limited to certain AWS regions	Broad availability across many AWS regions
Setup	API Keys	IAM + Instance Setup
Development	API Integration	Notebooks + SDK (Jupyter, TensorFlow, PyTorch)
Testing	API Testing	Full ML Testing (Data validation, performance testing)
Deployment	API Endpoint (Serverless)	Multiple Options (API, batch, real-time, edge)